Notes on lecture 5

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1 Continuity of a function of 2 variables

Definition 1.0.1. Let $f : D \subseteq \mathbb{R}^2 \to \mathbb{R}$ be a function of 2 variables. Let $(a, b) \in D$. We say that f is **continuous** at (a, b) if $\lim_{(x,y)\to(a,b)} f(x, y)$ exists and equals $f(a, b)$. $\square$

Remark 1.0.2. Let $f : D \subseteq \mathbb{R}^2 \to \mathbb{R}$ be a function of 2 variables. Let $(a, b) \in D$. For checking the continuity of f at (a, b), we should first check whether or not the limit $\lim_{(x,y)\to(a,b)} f(x, y)$ exists. If the limit doesn't exist, then there is no further question about the continuity of f at (a, b). In this case, we are not going to proceed any further, and we say that f is not continuous at (a, b).
If the limit $\lim_{(x,y)\to(a,b)} f(x, y)$ exists, then we should first check whether or not $f(a, b)$ is well defined. If $f(a, b)$ is not well defined, then there is no further question about the continuity of f at (a, b). In this case too, we are not going to proceed any further, and we say that f is not continuous at (a, b).
If $f(a, b)$ is well defined, then we should check whether or not $\lim_{(x,y)\to(a,b)} f(x, y)$ equals $f(a, b)$. If it equals $f(a, b)$, then we say that f is continuous at (a, b). Otherwise, we say that f is not continuous at (a, b).

Example 1.0.3. Check the continuity of $f(x, y) = \frac{x^2-y^2}{x^2+y^2}$ at $(0, 0)$.
Observe that we had already seen in the previous lecture that

$$\lim_{(x,y)\to(0,0)} \frac{x^2 - y^2}{x^2 + y^2}$$

does not exist. So, f is not continuous at $(0, 0)$. Moreover, the function f here is not well defined at $(0, 0)$. $\square$

Example 1.0.4. Recall that previously, we had proved that

$$\lim_{(x,y)\to(0,0)} \frac{3x^2y}{x^2+y^2} \text{ exists and equals } 0.$$

But the function $\frac{3x^2y}{x^2+y^2}$ is not well defined at $(0,0)$. Now let us take a function f given by

$$f(x,y) = \begin{cases} \frac{3x^2y}{x^2+y^2} \text{ when } (x,y) \neq (0,0) \\ 0 \text{ when } (x,y) = (0,0) \end{cases}$$

We can easily see now that $\lim_{(x,y)\to(0,0)} f(x,y)$ exists and equals $f(0,0)$. So the function f is continuous at $(0,0)$. □

Remark 1.0.5. One can easily check that polynomial functions of any degree are always continuous at all points in $\mathbb{R}^2$.

Remark 1.0.6. If we consider a rational function (that is, a quotient of 2 polynomial functions), then it is always continuous at any point in the domain of definition. The domain of definition of a rational function is $\mathbb{R}^2 \setminus Z_D$, where Z_D denotes the set of all zeroes of the polynomial in the denominator.

Example 1.0.7. The domain of definition of the rational function $\frac{x^2-y^2}{x^2+y^2}$ is $\mathbb{R}^2 \setminus \{(0,0)\}$, and this rational function is continuous at every point in $\mathbb{R}^2 \setminus \{(0,0)\}$. □

Definition 1.0.8. Let $D \subseteq \mathbb{R}^2$. If a function f is continuous at all points of D, then we say that f is **continuous** on D. □

2 Partial derivatives

2.1 The heat conduction problem

Suppose we have taken a metallic bar and we are interested in the temperature distribution at all points on the metallic bar. See Figure 2.1.1 for an illustration.

Suppose first we dip the bar in icecold water. Then we take it out after a while. Then we know that at all points on the bar, the temperature is going to be 0 degree celcius. Now again, let us impart some temperature (say, 100 degree celcius) at the top of the bar. If we now take a cross section of the bar

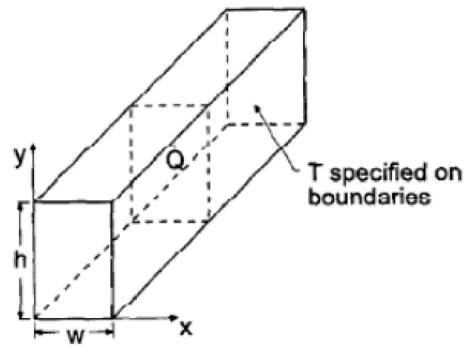

Figure 2.1.1: A metallic bar

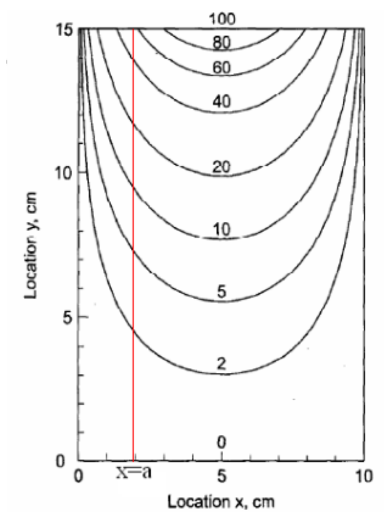

Figure 2.1.2: A cross section of the metallic bar

(by taking a slice through a vertical plane), then the cross section is going to look like a metallic plate as shown in Figure 2.1.2.

At the top, the temperature is going to be 100 degree celcius, and as you move from the top to the bottom of this cross section, the temperature is going to reduce gradually. At the bottom most point, the temperature is going to be 0 degree celcius.

After a while, the temperature inside the cross-sectional plate is going to settle down and the temperature distribution is going to look as in Figure 2.1.2.

The curves shown in Figure 2.1.2 are the isotherms.
From single variable calculus, we know that the derivative is nothing but the rate of change of something. So whenever we consider a function f of one independent variable x, we take some increment Δx and then we take the ratio $\frac{\Delta f}{\Delta x}$ for computing the derivative of f. That means, we try to find out the rate of change of f with respect to the single variable x.
Here, if we want to consider the rate of change of temperature (of the cross-sectional plate), we must have a direction along which we should consider the rate of change! For instance, we may consider the rate of change of temperature along the vertical direction or along the horizontal direction. Earlier, in single variable calculus, we were having only one direction to be considered for calculating the rate of change of a function. But here, we have infinitely many directions along which we can calculate the rate of change. One of these directions is the vertical direction, and another is the horizontal one. The vertical direction can be regarded as the direction along the y-axis. And the horizontal direction can be regarded as the direction along the x-axis.
If we consider the rate of change only along these 2 directions, that is, along the x and the y axes, then we are going to come up with the concept of the partial derivative. The rate of change along an arbitrary direction (not just along the x and the y axes), is going to the called the "directional derivative", which we will study later on.

2.2 Partial derivatives - Definition

Let f be a function of two variables x and y. Let (a, b) be a fixed point in the domain of f. Define a function $g_1(x)$ of the single variable x as follows:

$$g_1(x) := f(x, b).$$

Observe here that the y-coordinate of f is kept fixed as b. Then we have

$$g_1'(a) = \lim_{h \to 0} \frac{g_1(a+h) - g_1(a)}{h} = \lim_{h \to 0} \frac{f(a+h, b) - f(a, b)}{h}$$

This last limit $\lim_{h \to 0} \frac{f(a+h,b)-f(a,b)}{h}$ is defined as the **partial derivative of f with respect to x at** (a, b). It is denoted by $f_x(a, b)$.
Basically, although it is an ordinary derivative of the function $g_1(x)$ at a, we call it a partial derivative because we are actually considering a function of

2 variables.
Similarly, if we define a function $g_2(y)$ of the single variable y as

$$g_2(y) := f(a, y),$$

then $g_2'(b)$ will give us the **partial derivative of f with respect to y at (a,b)**. It is denoted by $f_y(a,b)$. That is, we have

$$g_2'(b) = \lim_{k\to 0} \frac{g_2(b+k) - g_2(b)}{k} = \lim_{k\to 0} \frac{f(a,b+k) - f(a,b)}{k} = f_y(a,b).$$

Till now, we were working with a fixed point (a,b) in the domain of f. But if we want to take a variable point (x,y) in the domain, then we have

$$f_x(x,y) = \lim_{h\to 0} \frac{f(x+h,y) - f(x,y)}{h}$$

and

$$f_y(x,y) = \lim_{k\to 0} \frac{f(x,y+k) - f(x,y)}{k}.$$

Sometimes the symbols $\frac{\partial f}{\partial x}$ and $D_x(x,y)$ are also used to denote $f_x(x,y)$. Similarly, the symbols $\frac{\partial f}{\partial y}$ and $D_y(x,y)$ are also used to denote $f_y(x,y)$.
Hence to find partial derivative of a function of 2 variables, we have to treat one among the two variables as a constant, and differentiate with respect to the other variable.

Example 2.2.1. Let $f(x,y) = x^4y^3 + \sin xy$. Find the partial derivatives of f.
To find f_x, we simply consider the variable y (in $f(x,y)$) as a constant, and differentiate with respect to x. Therefore,

$$f_x(x,y) = 4x^3y^3 + y\cos xy.$$

Similarly,

$$f_y(x,y) = 3x^4y^2 + x\cos xy.$$

$\square$

2.3 Geometrical interpretation of partial derivatives

Let f be a function of 2 variables x and y. Let (a, b) be a fixed point in the domain of f.
To find $f_x(a, b)$, we are going to fix b, that is, $y = b$. Now the equation $y = b$ represents a vertical plane and this plane cuts the surface corresponding to the graph of f in a curve (say, C_1). See Figure 2.3.1 for an illustration.

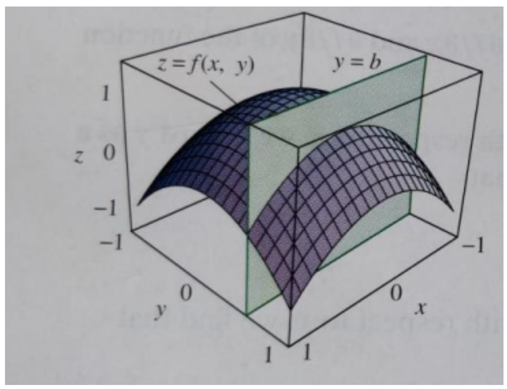

Figure 2.3.1: The surface $z = f(x, y)$ being cut by the plane $y = b$

Observe that the equation of the curve C_1 is given by $z = f(x, b)$. Figure 2.3.2 gives a magnified view of the phenomenon.
Let $g_1(x) = f(x, b)$. Then we know that

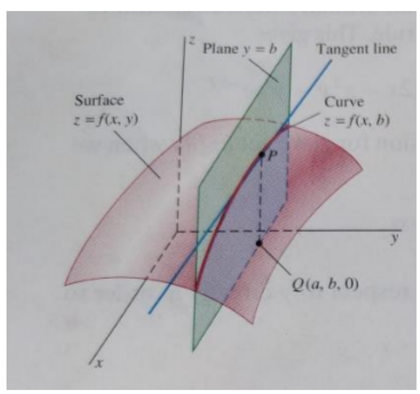

Figure 2.3.2: The curve $z = f(x, b)$

$$f_x(a,b) = g_1'(a) = \frac{dg_1}{dx}_{|x=a}.$$

But we know from single variable calculus that $\frac{dg_1}{dx}_{|x=a}$ is the slope of the tangent to the curve C_1 at the point (a,c), where $c = f(a,b)$. See Figure 2.3.3 for an illustration.

So geometrically, the partial derivative $f_x(a,b)$ of f at the point (a,b)

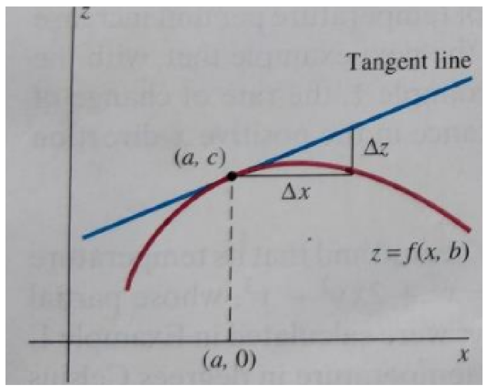

Figure 2.3.3: The tangent to the curve $z = f(x,b)$

signifies the slope of the tangent line at the point (a,c) to the curve cut out by the plane $y = b$ from the surface $z = f(x,y)$, where $c = f(a,b)$.
We can have a similar geometrical interpretation for $f_y(a,b)$.

2.4 Notation for partial derivatives

If $z = f(x,y)$, then we write

$$f_x(x,y) = f_x = \frac{\partial f}{\partial x}(x,y) = \frac{\partial z}{\partial x} = f_1 = D_1 f = D_x f.$$

Similarly, for $f_y(x,y)$ (For $f_y(x,y)$, we write f_2 and $D_2 f$).

2.5 Rule for finding partial derivatives

Let $z = f(x,y)$.

- To find f_x, regard y as a constant and differentiate $f(x,y)$ with respect to x.

- To find f_y, regard x as a constant and differentiate $f(x, y)$ with respect to y.

Exercise: Find the partial derivatives of the following functions:
(i) $f(x, y) = 3x - 2y^4$.
(ii) $f(x, y) = x^5 + 3x^2y^2 + 3xy^4$.
(iii) $f(x, y) = \int_y^x \cos(t^2)dt$.

3 Functions of more than 2 variables

Partial derivatives can also be defined for functions of 3 or more variables. For example, if f is a function of 3 variables x, y and z, then its partial derivative with respect to x is defined as

$$f_x(x, y, z) = \lim_{h \to 0} \frac{f(x + h, y, z) - f(x, y, z)}{h}.$$

It is found by treating y and z as constant and differentiating $f(x, y, z)$ with respect to x. If $w = f(x, y, z)$, then $f_x = \frac{\partial w}{\partial x}$ can be interpreted as the rate of change of w with respect to x when y and z are held fixed. But we can't interpret it geometrically because the graph of f lies in a 4-dimensional space.
In general, if u is a function of n variables, say $u = f(x_1, \dots, x_n)$, then its partial derivative with respect to the i-th variable x_i is given by

$$\frac{\partial u}{\partial x_i} = \lim_{h \to 0} \frac{f(x_1, \dots, x_{i-1}, x_i + h, x_{i+1}, \dots, x_n) - f(x_1, \dots, x_i, \dots, x_n)}{h}.$$

Observe that here we have all variables (except the i-th variable) fixed.
Let $x = (x_1, \dots, x_n) \in \mathbb{R}^n$. Write

$$\overline{x} =< x_1, \dots, x_n > .$$

Let $\hat{e_1}, \dots, \hat{e_n}$ denote the unit vectors along the directions $x_1, \dots, x_n$ respectively, where $\hat{e_i}$ is the vector in $\mathbb{R}^n$ having 1 at the i-th position and zeroes elsewhere. It is now easy to see that

$$\overline{x} + h\hat{e_i} =< x_1, \dots, x_{i-1}, x_i + h, x_{i+1}, \dots, x_n > .$$

We can then write

$$\frac{\partial u}{\partial x_i} = \lim_{h \to 0} \frac{f(\overline{x} + h\hat{e_i}) - f(\overline{x})}{h}.$$

4 Higher order partial derivatives

If f is a function of 2 variables x and y, then so are the partial derivatives f_x and f_y. So we can consider their partial derivatives $(f_x)_x, (f_x)_y, (f_y)_x, (f_y)_y$, which are called the **second order partial derivatives** of f. If $z = f(x, y)$, then we use the notation:

$$(f_x)_x = f_{xx} = f_{11} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 z}{\partial x^2}$$

$$(f_x)_y = f_{xy} = f_{12} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y\,\partial x} = \frac{\partial^2 z}{\partial y\,\partial x}$$

$$(f_y)_x = f_{yx} = f_{21} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial x\,\partial y} = \frac{\partial^2 z}{\partial x\,\partial y}$$

$$(f_y)_y = f_{yy} = f_{22} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial y^2} = \frac{\partial^2 z}{\partial y^2}$$

Thus the notation f_{xy} (or $\partial^2 f/\partial y\,\partial x$) means that we first differentiate with respect to x and then with respect to y, whereas in computing f_{yx} the order is reversed.

These are called **mixed partial derivatives**.

Theorem 4.0.1. ***Clairaut's theorem:*** *Suppose f is defined on a disc D that contains the point (a, b). If the functions f_{xy} and f_{yx} are both continuous on D, then $f_{xy}(a, b) = f_{yx}(a, b)$.*

Remark 4.0.2. Sometimes, directly differentiating a function with respect to x or y partially may not give you a correct picture. For some particular cases, we may need to find the partial derivative through the definition given by

$$f_x = \lim_{h\to 0} \frac{f(x + h, y) - f(x, y)}{h}.$$

Similar definition for f_y.
Sometimes, we may have to use the following definition (or similar definitions):

$$(f_x)_y = \lim_{k\to 0} \frac{f_x(x, y + k) - f_x(x, y)}{k}.$$

Example 4.0.3. Let $f(x,y) = x^4y^3 + \sin xy$. Then

$$f_x = 4x^3y^3 + y\cos xy.$$

$$f_{xx} = (f_x)_x = 12x^2y^3 - y^2\sin xy.$$

$$\text{and } f_{xy} = (f_x)_y = 12x^3y^2 - xy\sin xy + \cos xy.$$

Similarly, we can find f_y, f_{yx} and f_{yy}. Clearly for this example, Clairaut's theorem holds true.
But Clairaut's theorem may not hold true in all examples. □

5 Tangent planes

Let f be a function of 2 variables x and y. Let $z = f(x,y)$. Let (x_0, y_0) be a fixed point in the domain of f. Let $z_0 = f(x_0, y_0)$. Then the surface corresponding to $z = f(x,y)$ passes through the point $P = (x_0, y_0, z_0)$. Let us consider the tangent plane to this surface at the point (x_0, y_0, z_0).
If we know the direction ratios of the normal (say, $\vec{n} =< a, b, c >$) at the point (x_0, y_0, z_0) to the surface, then we can write down the equation of the tangent plane to the surface at the point (x_0, y_0, z_0) as follows:

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0.$$

If $c \neq 0$, then the above equation can be rewriiten as

$$z - z_0 = A(x - x_0) + B(y - y_0), \qquad (*)$$

where $A = -\frac{a}{c}$ and $B = -\frac{b}{c}$.
Now consider the plane $y = y_0$. Then this plane $y = y_0$ will cut out a curve (say, C_1) from the surface. And then, at the point $P = (x_0, y_0, z_0)$, we will have a tangent line (say, T_1). The partial derivative $f_x(x_0, y_0)$ is going to be the slope of the tangent line T_1. The equation $(*)$ along the plane $y = y_0$ reduces to

$$z - z_0 = A(x - x_0). \qquad (**)$$

Now equation $(**)$ also represents the same tangent line T_1, whose slope is $f_x(x_0, y_0)$. Hence we have

$$A = f_x(x_0, y_0).$$

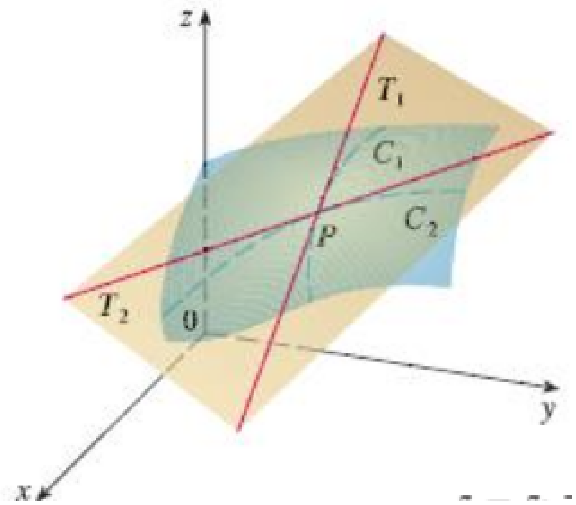

Figure 5.0.1: The tangent lines T_1 and T_2

Similarly, if we consider the plane $x = x_0$, then we will get a curve C_2, a tangent line T_2, and

$$B = f_y(x_0, y_0).$$

Figure 5.0.1 explains the phenomenon.
That means, if we want to find out the equation of the tangent plane to the surface $z = f(x, y)$ at the point (x_0, y_0, z_0) (where $z_0 = f(x_0, y_0)$), then we will have to simply find out $f_x(x_0, y_0)$ and $f_y(x_0, y_0)$, and replace A and B in equation $(*)$ by $f_x(x_0, y_0)$ and $f_y(x_0, y_0)$ respectively. So the **equation of the tangent plane** to the surface $z = f(x, y)$ at the point (x_0, y_0, z_0) is given by:

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0). \qquad (+)$$

Example 5.0.1. Write an equation of the tangent plane to the paraboloid $z = 5 - 2x^2 - y^2$ at the point $P = (1, 1, 2)$. See Figure 5.0.2 for an illustration.
Here, $z = f(x, y)$, where $f(x, y) = 5 - 2x^2 - y^2$. Also, $(x_0, y_0, z_0) = (1, 1, 2)$. So

$$f_x(1, 1) = -4 \text{ and } f_y(1, 1) = -2.$$

Plugging all this data in equation $(+)$, we get that the required equation of the tangent plane is given by

$$z - 2 = -4(x - 1) + (-2)(y - 1).$$

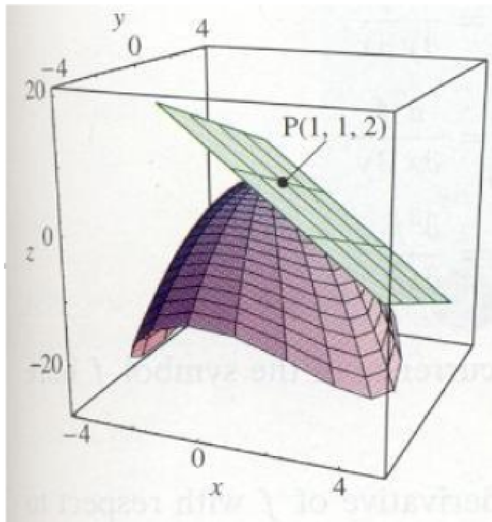

Figure 5.0.2: Figure for Example 5.0.1

Or in other words,

$$4x + 2y + z = 8.$$

$\square$

Remark 5.0.2. We know that the equation of the tangent plane to the surface $z = f(x, y)$ at the point (x_0, y_0, z_0) is given by

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

This equation can be rewriiten as

$$f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) + (-1)(z - z_0) = 0.$$

Now, $a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$ is also the equation of the same tangent plane, where $\vec{n} =< a, b, c >$ is the normal to the surface at the point (x_0, y_0, z_0).
If we compare the above two equations, then we get

$$< f_x(x_0, y_0), f_y(x_0, y_0), -1 >= \lambda < a, b, c >$$

for some scalar λ. This implies that $< f_x(x_0, y_0), f_y(x_0, y_0), -1 >$ is also going to give us the components of the normal to the surface at the point (x_0, y_0, z_0).

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